PERSONAL STUDY GUIDE | 12 NOVEMBER 2026

5G PHY & Wireless Baseband Development

Prepared for Dwikul Jyoti Das
A research-to-implementation preparation plan for wireless industry interviews and practical development work.

Your starting point

This guide uses your previously shared background: doctoral research at IIT Roorkee, M.Tech at IIT Jodhpur, B.Tech at NIT Silchar, MATLAB/Python experience, and application-development experience at Accenture during 2014-2016. It does not assume your current degree-completion status or a demonstrated production C/C++ skill level.

Your research in OFDM, OTFS, FBMC, MIMO, channel estimation, Doppler/CFO and optimization gives you substantial analytical preparation. The highest-value next step is to demonstrate how you turn those ideas into reproducible, standards-grounded receiver software with measurable correctness, latency and numerical behavior.

Role focus

Track	Preparation emphasis
PHY algorithm / modem systems	Receiver algorithms, estimation, synchronization, link-level evaluation and implementation trade-offs. Closest match to your research background.
L1 / baseband software developer	C/C++, NR processing chains, memory ownership, concurrency, tests, profiling and PHY-MAC integration. Main target of this guide.
Embedded DSP / firmware	Additional depth in fixed point, SIMD, hardware interfaces, interrupts, DMA and real-time deadlines. Add after the core track.

How to use the plan

Allow approximately 12 weeks at 15 hours/week. This is a planning estimate, not a guarantee of interview readiness. If C/C++ is new, extend the plan to 16-20 weeks and keep the project gates unchanged. At 7-8 hours/week, roughly double the calendar duration.

Suggested weekly allocation: 6 hours coding and debugging; 4 hours NR/specification study; 3 hours DSP and receiver derivations; 2 hours verbal interviews and revision. Your existing theory strength justifies concentrating effort on implementation.

Study loop: explain the principle, derive the essential relation, implement it, inject a failure, measure the result, then explain the trade-off in two minutes.

01 | EXECUTION PLAN

Twelve weeks, with evidence

Week	Focus	Deliverable / exit check
1	C/C++ baseline + DSP audit	Build a complex dot product and FIR filter; verify against MATLAB/Python; explain indexing and conjugation.
2	OFDM foundations	Noiseless and AWGN QPSK-OFDM link; validate FFT scaling, CP and noise variance.
3	NR grid and waveform	Generate a documented NR resource grid; explain numerology, PRBs, slots, DM-RS and payload overhead.
4	Synchronization	Estimate timing/CFO; plot residual CFO and detection failures versus SNR.
5	Channel estimation + MIMO	Compare LS/MMSE estimation and ZF/MMSE detection using matched assumptions and power normalization.
6	Coding and soft information	Use verified LDPC processing; check CRC, rate matching, LLR sign and decoder noise convention.
7	NR shared-channel chain	Run a constrained PDSCH/DL-SCH baseline; obtain BLER and goodput curves.
8	Control + uplink + HARQ	Trace PDCCH/PUCCH/PRACH/PUSCH roles; implement a small HARQ bookkeeping/combining experiment.
9	C/C++ port	Port one receiver kernel; compare intermediate arrays and end-to-end decisions to the reference.
10	Fixed point + performance	Quantize one kernel; report saturation, EVM/BLER change, latency distribution and memory use.
11	Integration + fault isolation	Trace one open-source PHY function through its callers and tests; document three injected faults.
12	Interview + portfolio	Run two timed mock interviews; finalize reproducible project instructions and a five-minute demo.

The first three study sessions

Session 1: implement complex multiply-accumulate with explicit conjugation and randomized test vectors. Compile with strong warnings and run address/undefined-behavior sanitizers where supported.

Session 2: generate QPSK symbols, perform IFFT/CP insertion, remove CP and FFT. Confirm noiseless recovery and Parseval consistency under your chosen normalization.

Session 3: add AWGN with a stated complex-sample variance; compare uncoded QPSK BER with theory. Write down why measured sample SNR differs from information-bit Eb/N0 once coding, inactive subcarriers and CP are included.

02 | FOUNDATION REFRESH

DSP you must explain and code

Study selectively, at implementation depth

Sampling and filtering: aliasing, anti-alias filtering, convolution versus correlation, FIR/IIR stability, group delay, decimation/interpolation and polyphase ideas. Implement a streaming FIR whose output remains identical across different input chunk sizes.

Transforms and complex baseband: DFT/FFT sign and scaling, fftshift versus physical subcarrier ordering, complex Gaussian noise, matched filtering and power spectral density. Understand when conjugation belongs in an inner product.

Digital communications: ML/MAP detection, Gray labeling, QPSK/QAM power normalization, BER versus SER, soft decisions, coding gain, Eb/N0 versus Es/N0, and diversity versus multiplexing.

Relations to derive without memorized shortcuts

OFDM: subcarrier spacing = Fs/NFFT; useful symbol duration = 1/subcarrier spacing. A sufficiently long CP and suitable FFT window convert a time-invariant finite multipath channel into circular convolution over the useful block. Significant within-symbol channel variation breaks the simple diagonal model.

Complex AWGN: for n = sqrt(sigma2/2)(a + jb), with independent unit-variance real Gaussian a and b, E[|n|^2] = sigma2. Define whether sigma2 is per complex time sample, FFT bin or receive antenna before using it.

Energy conversion: at the modulation-symbol level, Es/N0 = R log2(M) Eb/N0 when Eb is per information bit and R is the effective code rate. CP, pilots and inactive carriers require additional accounting when converting waveform sample SNR.

QPSK check: coherent uncoded Gray QPSK in AWGN has bit error probability Q(sqrt(2 Eb/N0)). Use this to expose normalization errors before introducing coding or fading.

Hands-on checks

Test an impulse through your FIR; a single active subcarrier through OFDM; a constant-phase rotation through the demapper; and a known delay through your synchronizer. Verify energy at every scaling boundary. Small deterministic vectors often expose bugs faster than a Monte Carlo curve.

Worked interview calculation

With Fs = 30.72 MHz and NFFT = 1024, subcarrier spacing is 30 kHz and useful symbol duration is about 33.33 microseconds. This does not determine the complete occupied bandwidth, CP pattern or NR carrier configuration: those need separate choices.

Exit check: reproduce a BER curve, explain every normalization, and localize a deliberately introduced factor-of-two noise error without rewriting the receiver.

03 | STANDARDS LITERACY

Learn NR through signal chains

Use a consistent set of specification versions for your project and record the release/version. Start with baseline NR features, then add release-specific features when a target job requires them. Read specifications by an implementation question rather than cover to cover. Specification roles are listed in [1].

Reference	Read it to answer
TS 38.211	Which physical signals/channels exist, and how are symbols and reference signals generated and mapped?
TS 38.212	How do CRC, code-block processing, coding and rate matching transform bits?
TS 38.213	What control-related PHY procedures and timing rules apply?
TS 38.214	How are data transmission parameters, MCS/TBS and related procedures determined?
TS 38.215	How are physical-layer measurements defined?
TS 38.300	How does NR fit into the wider radio-access architecture?

High-priority NR concepts

Numerology, normal versus extended CP, frames/subframes/slots, resource elements, PRBs, bandwidth parts, duplexing and TDD patterns; modulation and layer mapping; antenna ports versus physical antennas; DM-RS, PT-RS, CSI-RS and SRS. A PRB spans 12 subcarriers. For normal CP, a slot has 14 OFDM symbols; do not generalize this to every CP configuration.

For numerology index mu, subcarrier spacing is 15 x 2^mu kHz and normal-CP slot duration is 1/2^mu ms. At 30 kHz, a slot is 0.5 ms; the actual processing deadline is determined by scheduling, buffering and interfaces, not simply this slot duration.

Two chains you should draw from memory

Downlink shared data: transport block -> CRC and code-block processing -> LDPC/rate matching -> scrambling -> modulation -> layer mapping/precoding -> resource mapping -> OFDM. Receiver processing reverses the appropriate operations, adds synchronization/estimation/equalization, and ends with decoding and CRC.

Cell access: PSS/SSS detection and synchronization -> PBCH/MIB -> configured control/data reception for further system information -> random access procedures. Know where SSB, PDCCH, PDSCH, PRACH, PUSCH and PUCCH enter; random access is a procedure spanning PHY and MAC.

Exit check: select one legal single-layer configuration, explain its occupied resources and reference signals, and locate each relevant rule in the pinned specification set. Distinguish a generic OFDM demonstration from an NR-conformant implementation.

04 | YOUR TECHNICAL ADVANTAGE

Receiver algorithms and impairments

Turn your research knowledge into receiver decisions

Synchronization: timing acquisition versus tracking; integer/fractional CFO; sampling-frequency offset; phase noise; coarse/fine estimation and phase tracking. Inject impairments individually before combining them. Separate oscillator offset from Doppler caused by mobility and multipath geometry.

A normalized CFO epsilon = f_offset/subcarrier_spacing creates a time-domain rotation exp(j 2 pi epsilon n/NFFT). Residual CFO causes common phase evolution and ICI. A delay lying within a valid CP/FFT-window region can mainly appear as a phase slope across subcarriers; outside that region, ISI/ICI can emerge.

Channel estimation: LS pilot estimates, pilot interpolation, MMSE estimation, noise estimation, covariance assumptions and channel aging. Under the model y = Xh + n, with independent zero-mean h and n of covariances Rh and Rn, the linear MMSE form is Rh X^H (X Rh X^H + Rn)^(-1)y. Gaussian assumptions make it the conditional-mean MMSE estimator.

MIMO detection: for y = Hx + n, white noise variance sigma_n^2 and independent symbols of energy Es, W_MMSE = (H^H H + sigma_n^2/Es I)^(-1)H^H. Implement a numerical solve, not an explicit inverse. Explain rank deficiency, conditioning, noise enhancement and the meaning of the regularization term.

One experiment that fits your profile

Compare perfect-CSI and pilot-based receivers over a Doppler sweep at fixed resource allocation, total transmit power and channel normalization. Measure BLER, channel-estimation NMSE and residual CFO. Then vary pilot density. Report whether the loss is estimation aging, residual synchronization error or intrinsic within-symbol interference; do not infer the cause from BLER alone.

Practical fault-isolation map

Symptom	First checks
Noiseless link fails	Bit order, scrambling seed, RE mapping, FFT convention, layer ordering and CRC processing.
High-SNR error floor	Residual CFO, pilot mismatch, clipping, LLR sign, rate recovery and channel variation.
MATLAB passes; C++ fails	Array layout, off-by-one indices, conjugation, precision, bounds and uninitialized buffers.
More antennas reduce performance	Power normalization, antenna/layer ordering, channel conditioning and noise covariance.

Soft information, FEC and HARQ

Learn coding as a complete interface

Study CRC attachment/checking; code-block segmentation; LDPC base graphs and lifting; rate matching/recovery; filler/null handling; puncturing/repetition; iterative decoding and stopping rules. NR shared data uses LDPC. Study Polar processing for PBCH/DCI; UCI coding depends on payload and channel context, so avoid saying all control information uses Polar codes. Consult the exact configuration in TS 38.212 [1].

LLR convention: define L(b) = log[P(b=0|y)/P(b=1|y)]. Under this convention positive values favor zero. A max-log demapper approximates the log likelihood by minimum-distance terms, with scaling dependent on the stated complex-noise variance and constellation normalization. Match the decoder convention explicitly.

Equalization changes the effective channel gain and noise/interference statistics. Feeding the original pre-equalization noise variance into every soft demapper can miscalibrate confidence. Inspect post-equalization SINR, LLR histograms and clipping behavior.

HARQ and link adaptation

Distinguish ACK/NACK, new-data indication, HARQ process ID, redundancy version and soft-buffer lifetime. Chase combining combines repeated coded information; incremental redundancy supplies different selected coded bits. Align recovered bit positions and LLR conventions before combining.

Understand CQI/PMI/RI roles, MCS selection, transport-block size, code rate, overhead and retransmission cost. MAC makes scheduling decisions; PHY supplies measurements and executes configured processing. Do not confuse a PHY calculation with scheduler policy.

Metrics that make a result credible

BLER: failed blocks divided by transmitted blocks, specifying first transmission or post-HARQ. **Goodput:** successfully delivered unique information bits divided by elapsed simulated time, including the resource/time cost of retransmissions in the denominator. Raw coded-bit rate is a different metric.

Log seeds, configuration, tested blocks, errors and confidence intervals. If zero errors occur in N independent trials, a useful approximate one-sided 95% upper error-probability bound is 3/N. Zero observed errors do not establish zero true BLER; correlated fading trials need additional care.

Implementation exercise

Start with a verified encoder/decoder rather than writing an NR LDPC decoder during the first week. Trace one block through every buffer. Test a noiseless round trip, a known bit flip, LLR sign inversion and two HARQ transmissions. Show precisely where each deliberately introduced fault becomes visible.

Exit check: explain how apparently good EVM can coexist with failed CRCs, and why stronger LLR magnitudes do not always improve decoding.

06 | DEVELOPER PREPARATION

C/C++, fixed point and real time

C and C++ topics, in order

Core correctness: pointers, arrays, lifetime, stack/heap, const-correctness, integer widths, signedness, shifts, alignment, endianness, buffer bounds and undefined behavior. Implement bit packing/unpacking with explicit byte order and test boundary cases.

Modern C++: RAII, ownership, references, containers, move semantics, templates at a practical level, and non-owning views appropriate to the selected language standard. Keep allocation out of hot processing loops where predictable latency is required.

Tooling: Linux command line, Git, compiler warnings, CMake/build systems, debugger breakpoints/backtraces, sanitizers, unit tests and profiling. Use Python for test-vector generation, log analysis and regression automation.

Concurrency: producer/consumer queues, mutexes, condition variables, atomics, races, deadlocks, false sharing and priority inversion. Understand why volatile is not a thread-synchronization primitive. Do not start with lock-free code before you can explain ownership and ordering.

Fixed-point exercise

Represent a signed 16-bit value as integer/2^15, giving range [-1, 1-2^-15]. Multiplication produces additional fractional bits and needs a sufficiently wide intermediate. Specify rounding, rescaling and saturation; test negative full scale and positive near-full-scale inputs.

Quantize a complex dot product or FIR kernel at several word lengths. Measure saturation counts, output error and downstream EVM/BLER. Choose accumulator width from the maximum input magnitude and number of terms, not from the input type alone.

Performance engineering

Measure before optimizing. Profile scalar code, then investigate contiguous buffers, cache behavior, vectorization/SIMD and reduced copying. Benchmark warmed-up runs on a named CPU with recorded compiler flags. Report median and tail latency, workload and numerical tolerances; a single average runtime is insufficient.

Distinguish simulation speed, kernel latency and a real-time processing deadline. Passing a desktop benchmark does not prove modem timing compliance. For embedded roles, extend into DMA, interrupts, cache coherence, memory barriers and hardware accelerator interfaces.

Integration awareness

Learn PHY-MAC request/indication flows and FAPI concepts. For RAN-side roles, add CU/DU/RU roles and the O-RAN 7.2x split. These are architecture-dependent boundaries; use an actual implementation diagram rather than assuming every stack uses the same split [3].

Exit check: produce a tested C/C++ kernel with explicit API, reference comparison, boundary tests and reproducible performance results.

07 | PORTFOLIO EVIDENCE

Build one link and one strong kernel

Project A: reproducible NR downlink receiver

Scope: choose a constrained single-layer PDSCH/DL-SCH configuration with documented SCS, bandwidth allocation, modulation, coding, DM-RS and channel assumptions. Begin with perfect timing/CSI, then practical channel estimation and synchronization. Use a consistent NR reference and version.

Stages: noiseless mapping checks -> AWGN -> multipath -> practical estimation -> Doppler/CFO -> optional 2x2 MIMO. Use verified reference coding blocks. Label any unsupported or simplified features clearly rather than calling a partial chain a complete modem.

Evidence: BLER/goodput versus SNR, ideal-versus-practical CSI gap, residual CFO, NMSE, test-vector comparisons and a short fault log. MATLAB 5G Toolbox provides a useful PDSCH reference if you have a license [2]. Without it, build the generic OFDM base and use a verified open-source implementation for NR-specific reference blocks.

Project B: a C/C++ receiver kernel

Port one well-defined block: complex dot product/MRC, pilot interpolation, QAM soft demapping, or a small MIMO detector. Start scalar and floating point. Add fixed point or SIMD only after the reference comparison passes.

Evidence: deterministic and randomized tests; documented tolerances; separate exact bit-level tests; representative input sizes; CPU/compiler details; runtime distribution; memory use; and the end-to-end cost of numerical changes. Record measured gains only.

Your differentiator: a mobility study within Project A

Use your CFO/Doppler expertise to explain the performance-latency trade-off of one receiver improvement, such as more frequent estimation or better tracking. Hold the comparison fair and show where the added computation is justified. OTFS/FBMC can support the research discussion after you establish the NR baseline.

Package the work so someone else can run it

Include a README, dependency/version list, configuration file, one-command test procedure, saved test vectors, plotting script, concise results report and known limitations. Separate your contributions from reused components and follow their licenses. An SDR is optional; simulation and kernel work can be completed first.

Application and interview positioning

Prepare three explanations of each project: 30 seconds for the problem and outcome; two minutes for design and evidence; ten minutes for derivation, implementation and failures. Describe prior Accenture experience as software experience, and distinguish it from any demonstrated embedded/PHY experience.

Select role seniority using the specific job requirements and your relevant implementation evidence. A PhD supports research depth; it does not by itself establish production firmware seniority.

Twelve questions with answer anchors

1. Why does the CP help OFDM equalization?
It enables circular-convolution processing when channel memory and timing fit within the protected interval, under suitable channel stationarity. It does not remove CFO or arbitrary time variation.

2. Why is ZF fragile in a deep fade?
It inverts small channel gains or singular values and amplifies noise. MMSE regularization trades residual interference against noise enhancement.

3. Why can increasing SNR fail to remove an error floor?
The dominant error may be residual interference, synchronization, clipping or a deterministic mapping/coding bug. Isolate those with noiseless and ideal-CSI tests.

4. What is the difference between Doppler and CFO?
Oscillator mismatch is a carrier offset; mobility creates propagation-dependent Doppler. Multiple paths can have different Dopplers, which one common correction cannot remove.

5. What is an antenna port?
A logical reference-signal/channel relationship used by the receiver, not necessarily a one-to-one physical antenna connector. Precoding can map ports/layers across hardware elements.

6. Why are hard decisions insufficient for a good decoder?
They discard reliability. Soft LLRs retain relative confidence, but require consistent signs, scaling and clipping.

7. What is the difference between SRS and DM-RS?
SRS supports uplink sounding; DM-RS supports demodulation of its associated physical channel. Their configuration and intended uses differ.

8. Why can a MATLAB-to-C++ port change results?
Indexing, memory order, type conversion, conjugation, random sequences, FFT scaling and precision can differ. Compare intermediate buffers using shared vectors.

9. Why avoid explicit matrix inversion?
A linear solve generally offers better numerical and computational behavior. Use the matrix structure when valid, and check conditioning.

10. Why is volatile insufficient for thread safety?
It does not provide the required atomicity or inter-thread ordering. Use synchronization appropriate to the shared data and access pattern.

11. Does zero measured BLER prove error-free operation?
No. Report sample size and uncertainty. Under independent trials the approximate 95% upper bound after zero failures is 3/N.

12. How would you investigate a slot deadline miss?
Measure stage timings and tails; inspect queueing, allocations, contention, cache misses and I/O. Correlate with workload before changing algorithms.

Know when you can demonstrate it

Practical self-assessment

Score each area 0-3: 0 = unfamiliar; 1 = can explain; 2 = can derive and implement with references; 3 = can debug and defend trade-offs. Treat this as a study rubric, not an employer hiring rule.

Area	Evidence expected
DSP / OFDM	Correct reference link and explicit normalization; explain CP, CFO and FFT conventions.
NR literacy	Trace a PDSCH chain and locate relevant clauses in the pinned specs.
Receiver design	Compare estimation/detection choices under controlled assumptions.
Coding / HARQ	Trace bits and LLRs; explain rate recovery, combining and CRC outcomes.
C/C++	Tested kernel, safe buffers, ownership model, debugger and sanitizer use.
Numerics / timing	Quantization evidence and reproducible latency profiling.
Communication	Five-minute project demo plus three specific debugging stories.

Suggested application gate: no zero in DSP, NR, receiver design or C/C++; at least level 2 in those four areas; and one reproducible project. For firmware-heavy roles, require additional fixed-point and real-time evidence. Continue applications and learning together once you can demonstrate this baseline.

Use your existing interview bank

Tag questions by DSP, communications, NR, programming and debugging. For each, keep a short answer, derivation, code exercise and common trap. Revisit failed questions after one day, one week and one month. Prioritize weak implementation topics over repeatedly rehearsing familiar thesis material.

Sources and reading route

The curriculum, time allocation and readiness gates are tailored recommendations. The sources below support standards navigation and practical reference implementations; they are not promises about any particular employer's interview.

[1] 3GPP specifications portal
Open official source - Find TS 38.211-38.215 and 38.300. Record exact versions used; verify parameter-dependent rules in the documents.

[2] MathWorks: NR PDSCH Throughput
Open official source - Reference for practical/perfect synchronization and estimation, channel modeling and throughput simulation. Toolbox/license availability is required.

[3] srsRAN: O-RAN gNB Overview
Open official source - Read to connect PHY processing to CU/DU/RU boundaries and interfaces; check the version of the implementation you use.

[4] srsRAN downloads
Open official source - Entry point for software and documentation. Select the appropriate project and inspect its tests, supported features and licensing.

First milestone: a correct QPSK-OFDM reference link and one tested C/C++ kernel. Build the more elaborate NR and mobility work on that evidence.